

Julius Zhang

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I am interested in identifying structures and developing new primitives for practical problems.

EXPERIENCE

Columbia University - Conduct research in the Department of Mathematics, hosted by Andrew Blumberg. - Homotopy theory, arithmetic geometry, cryptography, security, and other topics.	Researcher, Jun. 2026 - Present
a16z - Building Jolt, open-source zero-knowledge virtual machine in Rust. - Implemented and benchmarked torus compression; proved optimality with a conceptual reframing via Hilbert 90. - Audited proof systems security: verify correctness of elliptic-curve optimizations and formalize folklore results into rigorous proofs. - Evaluated lattice-based cryptography parameters against known attacks under various security regimes and performance constraints. - Performance engineering across the proving stack, including sum-check optimization.	Researcher, Apr. 2025 - Present
Millennium Convex optimization and robust statistical estimation for real-time trading systems.	Researcher, Jun. 2024 - Apr. 2025
Stellar Development Foundation - Byzantine fault tolerance, C++, stellar-core. - Improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.	Consultant, Jun. 2023 - Sept. 2023
Distributed Systems and TCS Research, Stanford University - Under David Mazières, research in general open membership Byzantine fault tolerance protocols (Github), with applications to certificate authority management and automatic password recovery with hardware security modules. - Research PCP under Aviad Rubinfeld, reconstructed proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.	Researcher, Sept. 2023 - Jun. 2024
Topology Research, Stanford University - Floer homotopy theory under Mohammed Abouzaid; related the complex cobordism Gysin map to the Grothendieck residue symbol and derived a recursive formula for Gromov–Witten quantum product corrections. - Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincare conjecture.	Researcher, Jul. 2020 - May 2022

SELECTED PAPERS AND MANUSCRIPTS

Julius Zhang. <i>The Relative Trace-Zero Subgroup of the Barreto-Naehrig Curves</i> , Cryptology ePrint Archive, 2026/1311. [PDF]
Julius Zhang. <i>Complex Cobordism and Formal Group Law</i> , Manuscript (Undergraduate Honours Thesis), Stanford University, 2024. [PDF]

TALKS

- p -Ordinary Cohomology for $SL(2)$ over Number Fields**, Hida Theory Seminar, Columbia University, 2026.
- Construction of Seiberg-Witten-Floer stable homotopy types**, Homotopy Theory and Manifolds Seminar, Columbia University, 2026.

EDUCATION

Courant Institute of Mathematical Sciences, New York University , computer science <i>PhD</i> . 2026 Verifiable systems and theoretical computer science.	starting Sept.
Stanford University , computer science <i>M.S.</i> and mathematics <i>B.A.</i> General Wang Yaowu Scholarship.	Sept. 2019 - Jun. 2024

SKILLS

Computer Science: Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.

Mathematics: number theory, algebraic geometry, symplectic topology, homotopy theory, Floer theory, gauge theory, concentration inequalities.